Discussion Section 3
STAT 132 , Winter 2025

1. Suppose that $X_1, \ldots, X_n$ are a random sample from the Normal distribution with unknown mean μ and unknown variance σ^2. Find the maximum likelihood estimator for μ and σ^2?

2. A random variable X is modeled using the following PDF:

$$f(x|\theta) = \frac{\theta}{x^{\theta+1}} \quad \text{for } x > 1,$$

where $\theta > 0$. Suppose you observe the random sample $X_1, \ldots, X_n$. What is the MLE of θ?

3. Suppose that $X_1, \ldots, X_n$ form a random sample from the uniform distribution on the interval $[\theta_1, \theta_2]$, where both θ_1 and θ_2 are unknown $(-\infty < \theta_1 < \theta_2 < \infty)$. Find the MLE of θ_1 and θ_2.

4. Suppose that $X_1, \ldots, X_n$ form a random sample from an exponential distribution for which the value of the parameter λ is unknown. Determine the MLE of the median of the distribution.

5. Suppose that $X_1, \ldots, X_n$ are a random sample from the Normal distribution with unknown mean μ and unknown variance σ^2. Find the MLE of the 0.95 quantile of the distribution (i.e. the value of a such that $\mathbb{P}(X < a) = 0.95$).

$$\log L = -n \cdot \log\sqrt{2\pi} - n\log\sigma - \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{2\sigma^2}$$

$$\frac{\partial \log L}{\partial \sigma} = -\frac{n}{\sigma} + \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\sigma^3} = 0$$

$$\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\sigma^3} = \frac{n}{\sigma}$$

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n}(X_i-\hat{\mu})^2}{n} \qquad \hat{\mu} = \frac{\sum_{i=1}^{n} X_i}{n}$$

2. $L = \prod_{i=1}^{n} f(x) = \frac{\theta^n}{X_1^{\theta+1} \cdot X_2^{\theta+1} \cdots X_n^{\theta+1}} = \frac{\theta^n}{(X_1 X_2 \cdots X_n)^{\theta+1}}$

$\log ab$
$\log a + \log b$

$$\log L = n \cdot \log\theta - (\theta+1)\sum_{i=1}^{n} \log X_i$$

$$\frac{d \log L}{d\theta} = \frac{n}{\theta} - \sum_{i=1}^{n} \log X_i = 0$$

$$\theta = \frac{n}{\sum_{i=1}^{n} \log X_i}$$

3. $f(x) = \frac{1}{\theta_2 - \theta_1} \quad \theta_1 \le X \le \theta_2$ ←

→ $X_1 \cdots X_n$

→ $L(\theta) = \prod_{i=1}^{n} f(x_i) = \left(\frac{1}{\theta_2-\theta_1}\right)^n \quad \theta_1 \le X_1, \cdots X_n \le \theta_2$

θ ✓

→ $L(\theta)$ X

↓ $\log L$ X

$\min \theta_2 = \max\{X_1 \cdots X_n\}$

$\max \theta_1 = \min\{X_1 \cdots X_n\}$

4. $X_1 \cdots X_n \sim Exp(\lambda)$

$f(x) = \lambda e^{-\lambda x}$

$L(\lambda) = \lambda^n e^{-\lambda(\Sigma x_i)}$

$\log L = n \cdot \log\lambda - \lambda \cdot \Sigma x_i$

$\frac{d \log L}{d\lambda} = \frac{n}{\lambda} - \Sigma x_i = 0$

MLE

$\hat{\lambda} = \frac{n}{\Sigma x_i}$

Median $\rightarrow F(x) = \frac{1}{2}$

"CDF"

$= 1 - e^{-\lambda x} = \frac{1}{2}$

$-\lambda x = -\log 2$

$-\lambda x = \log \frac{1}{2}$

$e^{-\lambda x} = \frac{1}{2}$

$X_{Median} = \frac{\log 2}{n/\Sigma x_i}$

$\int_{-\infty}^{X_{Median}} f(x)dx = \frac{1}{2}$

$X_{Median} = \frac{\log 2}{\lambda}$

5. From (1) we know that

$\hat{\mu} = \frac{\sum_{i=1}^{n} X_i}{n}$, $\hat{\sigma}^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n}$

95 quantile: $a: P(X < a) = 0.95$

$X \sim N(\mu, \sigma^2) \Rightarrow \frac{X-\mu}{\sigma} \sim N(0,1)$

$P\left(\frac{X-\mu}{\sigma} < \Phi(0.95)\right) = 0.95$

$P(X < \Phi(0.95) \cdot \sigma + \mu) = 0.95$

$\hat{\sigma}$ $\hat{\mu}$

$\therefore \hat{a} = \Phi(0.95) \cdot \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n}} + \frac{\sum_{i=1}^{n} X_i}{n}$

Monday!

1. ① $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$

$$L(X_1 \cdots X_n) = \prod_{i=1}^{n} f(X_i) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left\{-\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{2\sigma^2}\right\}$$

$$\log L = \log(X_1 \cdots X_n) = -n \cdot \log\left(\sqrt{2\pi\sigma^2}\right) - \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{2\sigma^2}$$

$$= -n\log(\sqrt{2\pi}) - n\log(\sigma) - \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{2\sigma^2}$$

$$\frac{\partial \log L}{\partial \mu} = \frac{\partial \sum_{i=1}^{n}(X_i-\mu)^2}{\partial \mu} = n\cdot\mu^2 - 2\cdot\sum X_i \cdot \mu + \cdots$$

① $\frac{\partial}{\partial} = 0$ $2n\mu - 2\sum X_i = 0$

$\mu = \frac{\sum X_i}{n}$

② $n\cdot\left(\mu^2 - 2\cdot\frac{\sum X_i}{n}\cdot\mu + \cdots\right)$

$n\cdot(\mu - \frac{\sum X_i}{n})^2 + \cdots$

≥ 0

$\mu = \frac{\sum X_i}{n}$

$(X_1-\mu)^2 + (X_2-\mu)^2 + \cdots (X_n-\mu)^2$

$X_1^2 - 2X_1\mu + \mu^2 + X_2^2 - 2X_2\mu + \mu^2 + \cdots X_n^2 - 2X_n\mu + \mu^2 = \sum X_i^2 + n\mu^2 - 2\sum X_i\mu$

$$-n\log(\sigma) - \frac{\sum(X_i-\mu)^2}{2\sigma^2}$$

MLE: $\frac{\sum X_i}{n} = \bar{X}$

$$\frac{\partial L}{\partial \sigma} = -n\cdot\frac{1}{\sigma} + \frac{\sum(X_i-\mu)^2}{\sigma^3}$$

$$= -\frac{n}{\sigma} + \frac{\sum(X_i-\bar{X})^2}{\sigma^3} = 0$$

$$\sigma^2 = \frac{1}{n}\sum(X_i-\bar{X})^2$$

2. $f(x) = \frac{\theta}{X^{\theta+1}}$ $X_1 \cdots X_n$ MLE for θ

$\log(ab) = \log a + \log b$

$$L = \prod_{i=1}^{n} f(x_i) = \frac{\theta^n}{X_1^{\theta+1} \cdot X_2^{\theta+1} \cdots X_n^{\theta+1}} = \frac{\theta^n}{(X_1 X_2 \cdots X_n)^{\theta+1}}$$

$$\log L = n \cdot \log\theta - (\theta+1) \cdot \log(X_1 X_2 X_3 \cdots X_n)$$

$$= n \cdot \log\theta - (\theta+1) \cdot \sum_{i=1}^{n} \log(X_i)$$

$$\frac{d \log L}{d\theta} = \frac{n}{\theta} - \sum_{i=1}^{n} \log(X_i) = 0$$

$$\text{MLE} \Rightarrow \theta = \frac{n}{\sum_{i=1}^{n} \log(X_i)}$$

3. $X_1 \cdots X_n \sim Unif(\theta_1, \theta_2)$

$f(x) = \frac{1}{\theta_2 - \theta_1}$, $\theta_1 \le X \le \theta_2$

$\Rightarrow L(\theta_1, \theta_2) = \prod_{i=1}^{n} f(x_i) = \frac{1}{(\theta_2 - \theta_1)^n}$ $\quad \theta_1 \le X_1 \cdots X_n \le \theta_2$

maximize ↑ ; minimize $\theta_2 - \theta_1$ ↓

$\theta_2 \downarrow$ $\quad \theta_1 \uparrow$

$\Rightarrow$ largest $X_i = X_{(n)}$ $\quad$ smallest $X_i = X_{(1)}$

4. $X_1 \cdots X_n \sim Exp(\lambda)$

$f(x) = \lambda e^{-\lambda x}$

$L(X_1 \cdots X_n) = \lambda^n e^{-\lambda \sum X_i}$

$\log L = n \cdot \log \lambda - \lambda \sum X_i$

$\frac{\partial \log L}{\partial \lambda} = \frac{n}{\lambda} - \sum X_i = 0$

$\lambda = \frac{n}{\sum X_i}$ ↑ MLE

Median: $F(X_m) = 0.5$

$F(X_m) \to Exp(\lambda) = 1 - e^{-\lambda X_m} = \frac{1}{2}$

$X_m = \frac{\log 2}{\lambda}$

$\hat{X}_m = \frac{\log 2}{\frac{n}{\sum X_i}}$ → MLE